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30 July 1981
SUPERSEDING
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5 May 1975

MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, POLYESTER AND RAYON, CREASE-RESISTANT FINISH

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a polyester and rayon cloth with a crease-resistant finish.

1.2 Classification. The cloth shall be one type in the following classes as specified (see 6.2):

Class 1 - Blue 3346
Class 2 - Blue 3362

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1133 Packaging and Packing of Synthetic Fiber Fabrics

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered
MIL-T-43624 - Thread, Polyester, Spun

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Application for copies should be addressed to Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and uniformity of shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 6.2 and 4.3).

3.3 Materials. (See 6.5).

3.3.1 Fiber blend.

3.3.1.1 Polyester fiber. The fiber shall be polyethylene glycol terephthalate when tested as specified in 4.5.

* 3.3.1.2 Rayon fiber. The fiber shall be a regular or high wet-modulus type rayon staple when tested as specified in 4.5. The use of rayon waste of any kind is prohibited.

* 3.3.2 Yarn. The warp and filling yarns shall be a blend of 65 (± 5.0) percent polyester and the remainder shall be rayon. As an alternate, a 50 (± 5.0) percent polyester blend is acceptable for the high wet-modulus type rayon fiber. The warp shall be a two-ply yarn and the filling can be either a two-ply or single yarn when tested as specified in 4.5.

* 3.4 Physical requirements. The finished fabric shall conform to the requirements in Table I when tested as specified in 4.5.

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Table I - Physical requirements

Weight oz/yd^2 (g/m^2)	Yarns per Inch (per 2.5 cm)		Breaking Strength (lbs) (Newtons)		Tearing Strengths (lbs.) (Newtons)		Weave
Minimum	Minimum		Minimum		Minimum		
	Warp	Filling	Warp	Filling	Warp	Filling	
5.7 (193.2)	50	40	120 (534)	90 (401)	7.5 (33)	6.5 (29)	Plain

3.5 Color. The color of the finished cloth shall be Blue 3346 for class 1, and Blue 3362 for class 2 cloth and shall match the standard sample.

3.5.1 Matching. The color shall match the approved standard shade under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.5.2 Colorfastness. The finished fabric shall show fastness to laundering, light, crocking and perspiration equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available, the finished cloth shall show a minimum of "good" fastness to laundering, light and perspiration and a Munsell Value of not less than 8.5 for crocking when tested as specified in 4.5.

3.5.3 Uniformity of shade. The uniformity of shade shall be equal to or better than that displayed by the standard sample.

3.5.4 Sulfur content. The use of sulfur dyes and dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When no standard sample is available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

3.6 Finish. The cloth shall be scoured, dyed, heat set, treated with a resin and cured to provide a crease-resistant finish and hand equal in characteristics to that of the standard sample.

3.7 Nonfibrous material. The total nonfibrous material content of resin finished cloth shall not exceed 8.5 percent when tested as specified in 4.5.

3.8 Width. The minimum width (inclusive of the selvage) of the finished cloth shall be as specified (see 6.2).

3.9 Seam efficiency. The cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.5.

3.10 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

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3.11 Dimensional stability. The finished cloth shall be preshrunk and shall not shrink or elongate more than 2.5 percent in either the warp or filling direction when tested as specified in 4.5. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket or package.

3.12 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards (45.7m). Each length shall be put up on full width rolls as specified in PPP-P-1133.

3.13 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act.

3.14 Marking. The face side of the cloth shall be identified by stamping on that side with the word "FACE" at each end of each piece.

3.15 Workmanship. The finished cloth shall conform to the quality established by this specifiation. The demerit points per 100 square yards ($83.6m^2$) when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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* 4.3 First article inspection. When required, the first article of finished cloth shall be visually inspected for uniformity of shade, color, finish, and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. All defects as defined in Section III of FED-STD-4, which are clearly visible at normal inspection distance (3 feet) (0.9m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "D" for slubs and "C" for knots, shall be scored. No linear yard (increments of 1 yard (0.9m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceed 38.0 points. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. If one or more rolls exceed the individual point value, the lot shall be rejected. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 Square yards } (83.6m^2)$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches (7.6 cm) or less in any dimension - one point

For defects exceeding 3 inches (7.6 cm) but not exceeding 6 inches (15.2 cm) in any dimension - two points

For defects exceeding 6 inches (15.2 cm) but not exceeding 9 inches (22.9 cm) in any dimension - three points

For defects exceeding 9 inches (22.9 cm) in any dimension - four points

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The following defects, when present, shall be scored four points for each yard (0.9m) in which they occur:

Baggy, ridgy or wavy cloth.
Width less than specified.
Characteristics of finish, hand and shade not equal to the standard sample.
Non-uniformity of shade (poor dye penetration, mottled, streaky or cloudy).
Excessive neppiness.

4.4.2.2 Examination for length

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

4.4.2.2.2 Total yardage in the sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade variation. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shade variation side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification, identification of preshrinking process and non-conformance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with the Section 5 requirements of this specification.

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4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II, shall be followed. The physical and chemical values specified in Section 3, except when otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 4-1/2 yards (4.1m) full width of the finished cloth. The lot size shall be expressed in units of 1 yard (0.9m). The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test methods

Characteristic	Requirement reference	Test method
Fiber identification	3.3.1	1/
Fiber content:		
Polyester (percent)	3.3.2	2100 and 4.5.1
Rayon (percent)	3.3.2	2100 and 4.5.1
Weight	3.4	5041
Yarns per inch (warp and filling)	3.4	5050
Breaking strength	3.4	5100
Tearing strength	3.4	5132
Weave	3.4	Visual 2/
Colorfastness to:		
Laundering	3.5.2	5614 3/
Light	3.5.2	5660
Perspiration	3.5.2	5680
Crocking	3.5.2	5651
Sulfur content	3.5.4	2020
Finish	3.6	1/
Nonfibrous material	3.7	2610
Seam efficiency	3.9	5110 4/
pH	3.10	2811
Dimensional stability	3.11	5556 5/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.
2/ One determination shall be made and the result reported as "pass" or "fail".
3/ Only the stain on the polyester and cotton fibers of the transfer cloth shall be evaluated.

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4/ The needle shall measure 0.040 inch (± 0.001 inch) across the blade at the eye. The thread shall be either polyester cotton-covered in accordance with MIL-T-43548, ticket No. 50, 2-ply for the needle and ticket No. 70, 2-ply for the looper or spun polyester in accordance with MIL-T-43624, class 1 or 2, ticket No. 40, 3-ply for the needle and ticket No. 70, 2-ply for the looper.

5/ The cotton laundering procedure shall be used.

4.5.1 Fiber content. The general procedures of test method 2100 shall be followed and the fiber content percentages shall be calculated as follows:

$$\frac{\text{Weight of dry residual fiber}}{\text{Weight of dry desized specimen}} \times 100 = \text{percent polyester}$$

100 - percent polyester = percent rayon

4.5.1.1 Report. Two specimens shall be tested from each sample unit and the average percent polyester and the average percent rayon of the two specimens shall be reported to the nearest 0.1 percent.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C (see 6.2).

5.1.1 Levels A and C. The cloth, put up as specified (see 3.12), shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The fabric covered by this specification is intended for use in shirts worn by the male personnel of the Navy and Coast Guard.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.2).
c. Class of cloth required (see 1.2).
d. Width required (see 3.8).
e. Minimum length if other than specified (see 3.12).
f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, "Sears Defect Replica Kit," Sears Tower, Chicago, IL 60684.

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification. Rayon waste of any kind is prohibited.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was due as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-29147A(NU) Cloth, Plain Weave, Polyester and Rayon, Crease-Resistant Finish

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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